PROJECT DESIGN PHASE
SOLUTION ARCHITECTURE

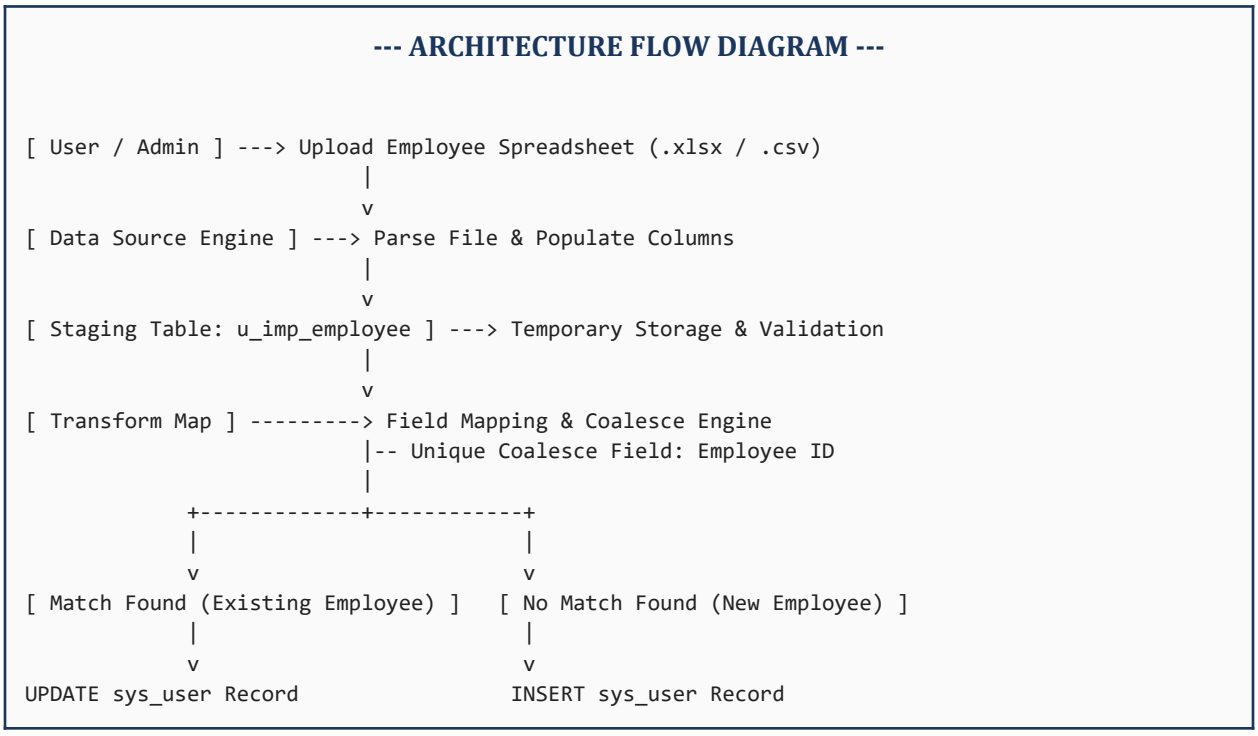
Team ID: SWTID-2026-1398	**Date:** 30 September 2026
Project Name: Import Data Using Transform Maps	**Maximum Marks:** 4 Marks

Solution Architecture Overview:

Solution architecture is a comprehensive process that bridges the gap between business requirements and technical implementation. Its core objectives are:

- Find the best technology solution to solve existing business and data integration challenges.
- Describe the structure, components, data flow, and behavior of the software system for project stakeholders.
- Define pipeline features, staging phases, transformation mappings, and technical system requirements.
- Provide explicit operational specifications according to which the solution is configured, validated, managed, and delivered.

Solution Architecture Diagram & System Workflow:



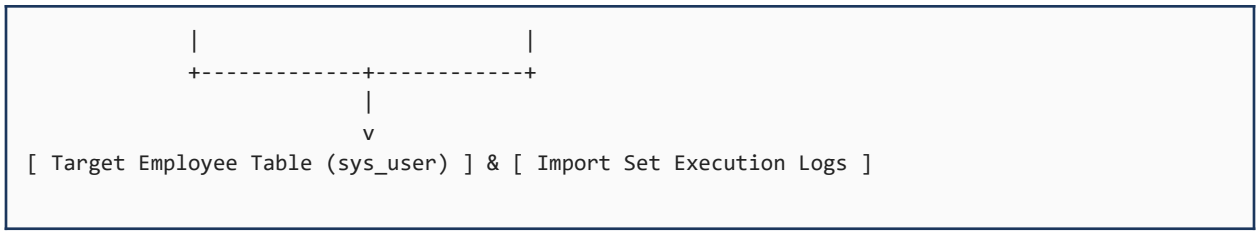


Figure 1: Automated Data Ingestion & Transformation Architecture in ServiceNow

Architectural Component Breakdown:

S.No.	Component	Architectural Function	ServiceNow Platform Technology
1	**Data Source / File Ingestion**	Accepts external XLSX/CSV spreadsheets containing employee attributes: Employee ID, Name, Email, Department, and Location.	ServiceNow Data Source Engine / System Attachment
2	**Import Set Staging Table**	Acts as an isolated staging table (`u_imp_employee`) holding raw imported rows prior to target commit, providing a validation buffer.	ServiceNow Import Set Staging Table
3	**Transform Map Engine**	Defines direct field-to-field relationships between staging columns and target table fields (`sys_user`), executing data transformations.	ServiceNow Transform Map (`sys_transform_map`)
4	**Coalesce Logic Engine**	Evaluates `Employee ID` as a unique primary key match. If a match exists, it updates the record; otherwise, it inserts a new employee profile.	ServiceNow Coalesce Field Mapping (`sys_transform_entry`)
5	**Target Master Database**	Stores active employee records with synchronized contact details, department allocations, and locations.	ServiceNow Employee Master Table (`sys_user`)
6	**Execution Logging & Audit**	Tracks transformation state (Inserted, Updated, Ignored, Error) for full transparency and operational auditing.	Import Set Run Logs (`sys_import_set_run`)

References:

- https://docs.servicenow.com/bundle/utah-platform-administration/page/administer/import-sets/concept/c_ImportDataUsingTransformMaps.html
- https://developer.servicenow.com/dev.do#!/learn/learning-plans/tokyo/servicenow_application_developer/app_store_learnv2_importingdata_tokyo_transform_maps
- https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/